

SALO KARKKA, Finland

WMO: 027560

Lat: 60.374N

Lon: 23.113E

Elev: 7

StdP: 14.69

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.4	-1.1	-12.8	2.8	-8.2	-5.7	4.1	-0.8	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.3	82.3	66.5	78.5	64.7	75.2	62.8	69.2	77.6	66.8	75.2	64.7	72.9	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.2	96.7	73.4	63.9	88.8	70.3	61.6	82.0	67.9	33.3	77.4	31.4	74.9	29.8	73.2	76.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
N/A	N/A	N/A	DB	-10.8	84.8	8.1	4.8	-16.6	88.3	-21.4	91.1	-25.9	93.8	-31.8	97.3
			WB	-11.2	71.0	8.0	3.4	-17.0	73.5	-21.7	75.5	-26.2	77.4	-32.0	79.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	42.7	22.1	24.9	29.8	40.4	51.3	58.5	64.0	60.8	52.8	41.7	36.0	29.0
	DBStd	15.97	11.16	10.61	8.30	5.78	7.58	5.82	5.45	4.98	5.67	6.56	7.68	9.55
	HDD50	3938	866	702	626	292	75	5	0	0	36	264	421	651
	HDD65	8233	1331	1122	1091	738	429	210	87	149	367	722	871	1116
	CDD50	1273	0	0	0	4	114	260	433	336	119	7	0	0
	CDD65	92	0	0	0	0	3	15	55	19	0	0	0	0
	CDH74	979	0	0	0	0	82	182	566	148	1	0	0	0
	CDH80	203	0	0	0	0	9	29	134	31	0	0	0	0
Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Precipitation	PrecAvg	25.9	2.2	1.7	1.3	1.3	1.3	2.2	2.8	3.2	2.0	2.7	2.6	2.4
	PrecMax	30.6	3.6	3.9	3.5	2.2	2.7	3.9	5.6	6.7	5.4	6.2	6.2	4.1
	PrecMin	21.3	0.3	0.0	0.1	0.1	0.2	0.4	0.2	0.4	0.4	0.5	0.1	0.2
	PrecStd	2.3	0.9	1.0	0.7	0.6	0.7	0.9	1.4	1.6	1.2	1.4	1.5	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.8	42.1	50.9	67.4	81.2	83.6	87.8	84.5	71.5	58.7	51.1	47.3
		MCWB	39.0	38.5	41.5	50.9	60.6	65.3	67.7	67.9	62.0	54.1	49.5	45.4
	2%	DB	38.9	39.6	46.2	61.6	75.9	79.2	83.7	77.6	67.5	55.5	49.3	44.1
		MCWB	37.7	37.3	39.9	48.3	58.4	62.7	67.5	65.2	59.9	52.6	48.1	42.4
	5%	DB	36.9	37.8	42.6	57.2	72.1	75.2	80.1	74.0	65.1	53.6	47.0	41.9
		MCWB	35.9	36.4	38.1	46.1	57.0	61.6	66.7	63.3	58.2	51.3	45.7	40.5
	10%	DB	35.1	36.3	39.6	52.8	67.7	71.3	76.5	71.2	62.6	51.9	45.4	39.9
		MCWB	34.2	35.2	36.1	43.4	55.1	58.8	65.4	61.7	57.7	49.8	44.2	38.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.6	39.6	43.2	52.2	64.2	68.4	72.7	70.0	63.2	55.7	50.0	46.2
		MCDB	40.6	40.9	48.0	63.8	75.5	77.7	81.5	78.4	68.2	57.3	50.7	46.8
	2%	WB	37.9	37.8	40.6	49.5	61.6	65.4	70.5	67.3	61.7	53.7	48.1	43.1
		MCDB	38.8	39.0	44.7	59.8	71.4	76.6	79.4	74.0	66.1	54.9	49.0	44.1
	5%	WB	36.2	36.6	38.7	47.0	58.6	63.0	68.4	65.2	60.0	51.9	46.0	40.8
		MCDB	36.9	37.7	41.9	55.8	69.0	72.6	76.8	71.6	63.7	53.1	46.8	41.9
	10%	WB	34.4	35.4	36.9	44.6	55.7	60.7	66.2	63.5	58.3	50.2	44.3	38.8
		MCDB	35.1	36.5	39.2	51.3	66.3	68.8	74.8	69.2	61.6	51.5	45.2	39.8
Mean Daily Temperature Range	5% DB	MDBR	10.7	11.1	14.4	18.4	20.4	17.7	18.3	17.1	15.3	12.6	7.7	8.6
		MCDBR	6.6	9.4	16.2	27.9	28.9	24.5	24.0	22.0	19.0	13.6	8.3	9.0
		MCWBR	6.4	8.0	10.8	16.1	14.1	11.9	10.7	11.3	12.0	10.7	7.9	8.5
	5% WB	MCDBR	6.9	8.3	13.9	24.3	24.6	21.3	20.4	17.5	16.1	11.0	8.5	8.7
		MCWBR	6.8	7.2	10.0	15.0	13.1	11.3	10.4	9.7	10.9	9.8	8.2	8.3
Clear-Sky Solar Irradiance	taub	0.273	0.310	0.333	0.372	0.367	0.361	0.388	0.380	0.350	0.332	0.302	N/A	
	taud	2.136	2.205	2.284	2.321	2.400	2.442	2.388	2.406	2.481	2.399	2.212	N/A	
	Ebn at Noon	150	213	250	261	272	276	264	257	246	207	138	N/A	
	Edh at Noon	13	23	29	33	33	32	33	30	24	19	13	N/A	
All-Sky Solar Radiation	RadAvg	77	256	732	1230	1681	1762	1660	1306	802	350	97	47	
	RadStd	14	51	89	124	166	139	162	137	101	56	18	5	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (59 neighbors)	+0.77	N/A	N/A	N/A	N/A	N/A	-243	N/A	N/A	N/A

Nomenclature: See separate page